

Academia- Industry Collaboration

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My Background

- IBM Research 8 years
- NV Research 9 years
- Collaborated with many universities: Berkeley, CUHK, Duke, GIT, Maryland, NCTU, NTU, PKU, TAMU, UCSD, UCLA, UIUC, UT Austin, Utah, etc

Diverse collaboration experience

- Industry leverages published ideas created by academics to solve their problems
- Industry collaborates with the academics to solve a new problem – funded or unfunded
- Industry hires interns from an academic group, part-time interns
- Industry/academic jointly apply for a research grant from the government
- Industry sponsors university research centers
- Industry provides supporting letters for academic grant applications
- Industry provides computing resources to academic
- Industry publishes problems (contest/benchmark) for academic to solve

Industrial Research Lab

- Industrial research looks for impact, not only novelty.
- Industrial research focuses on problems that can potentially impact its business: revenue or cost
- Industrial research lab can articulate the research questions better than industrial product teams
- What industrial research lab cares about academic collaboration
 - Expertise in a particular area that they are interested in
 - Strong student recruitment pipeline
 - Long-term relationship
 - IP issue – joined IP difficult, open source!

What lessons did you learn in your own academia-industry collaboration experience?

- What does a win-win look like?
- Project becomes deployed in products or released as a hot open-source repo, best paper award publications, and students recruited by the industry
- Ideas most important – both targeted problems and technical approaches
 - Niche problem, not well-defined problem, unsolvable problem
 - Lack of understanding of the problem domain, mundane technical approach, and lack of innovation
- “Rome wasn’t built in a day” – requires long-term commitment to a research direction
- “It takes two to tangle” - requires commitment from both sides

For young faculty/professionals, when is the best time to start academia-industry collaboration?

- For young faculty: perhaps after you have established a portfolio of research results and have some unique perspectives to offer to the industry
- For your professionals: when you can define a good problem direction but need extra help , and can identify an academic group who has expertise in that direction

What are the best ways to connect with industry/academia?

- Follow the publications
- Meet at conferences
- Reach out to each other for invited talks or discussions
- Organize regular meetings to discuss projects
- Work closely on projects
- Joint publications

Will the current federal funding situation change how academia-industry work together?

- Seen more academic requests for industrial funding
- But difficult for the industry to cover the federal funding deficit
- Academics are considering moving to industry jobs or startups
- Will Academic research be more applied and impact-driven?
- Collaboration becomes more important
- Be optimistic and think long term!